



LABORATORY JOURNAL (MP408)

Experimental Techniques – II Computational Methods in Physics

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Certificate

The Experiments Entered in This Journal Have Been Satisfactorily
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TABLE OF CONTENTS

SNo.	Date	Program	Page
P2.	10.02.2021	Practical 2	01
5.	10.02.2021	To write an Octave program for the animation of planetary motions of the solar system, incorporating Keplerian orbits, GR precession for Mercury, the Moon, and Halley's Comet.	02



Computational Methods in Physics (MP408)

GNU Octave

PRACTICAL – 2

Simulation of Planetary Motion

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PROGRAM – 5

Aim

To write an Octave program for producing a real-time **animation of planetary motion** in the solar system, incorporating Keplerian orbital mechanics for 7 planets, General Relativistic perihelion precession for Mercury, the Moon orbiting Earth, and Halley's Comet with high eccentricity ($e = 0.85$) and a rotated orbital plane.

Theory

The motion of a planet around the Sun is governed by Newton's law of gravitation. The force between two masses m_1 and m_2 separated by a distance r is:

$$\mathbf{F} = -\frac{Gm_1m_2}{r^3} \mathbf{r} \quad (1)$$

where $G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ is the universal gravitational constant. The negative sign indicates an attractive force.

If the two masses are located at $\mathbf{r}_1(x_1, y_1)$ and $\mathbf{r}_2(x_2, y_2)$, the distance between them is $r_{12} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. For two-dimensional motion in the xy -plane:

$$x(r, \theta) = r \cos \theta, \quad y(r, \theta) = r \sin \theta \quad (2)$$

Kepler's Laws:

1. *Law of Ellipses*: each planet orbits in an ellipse with the Sun at one focus. In polar form: $r(\theta) = a(1 - e^2)/(1 + e \cos \theta)$.
2. *Law of Equal Areas*: $r^2 \dot{\theta} = \text{const}$ (angular momentum conservation), implemented as $\Delta\theta \propto \Delta t/r^2$.
3. *Harmonic Law*: $T^2 \propto a^3$, so $\bar{\omega} \propto a^{-3/2}$.

Planet	a (AU)	e	$q = a(1 - e)$	$Q = a(1 + e)$	$\bar{v} \propto 1/\sqrt{a}$
Mercury	0.6	0.50	0.30	0.90	1.29
Venus	1.1	0.01	1.09	1.11	0.95
Earth	1.8	0.02	1.76	1.84	0.75
Mars	2.5	0.20	2.00	3.00	0.63
Jupiter	4.0	0.05	3.80	4.20	0.50
Saturn	5.5	0.06	5.17	5.83	0.43
Neptune	7.0	0.01	6.93	7.07	0.38

General Relativistic Precession (Mercury): GR predicts an additional perihelion advance of $43''/\text{century}$, unexplained by Newtonian mechanics. Modelled here as a continuous rotation of Mercury's position and orbit ellipse by $\Delta\omega = 0.4 \Delta t$ per frame.

Moon orbits Earth at display radius 0.3 AU with angular speed $10\times$ that of Earth. **Halley's Comet** ($e = 0.85$, $a = 5.5$, plane rotated $\pi/4$ rad) dramatically accelerates near perihelion, demonstrating Kepler's Second Law.

Program

The complete program is in P2.m (209 lines). Shown in full below:

Listing 1: Solar System Master Simulation — Setup, Legend Panel, Controls (P2.m)

```

1 % PH408: Practical 2 - Master Solar System (Complete Legend)
2 clear all; close all; clc;
3
4 % --- Configuration ---
5 if (ismember('qt', available_graphics_toolkits()))
6     graphics_toolkit('qt');
7 elseif (ismember('fltk', available_graphics_toolkits()))
8     graphics_toolkit('fltk');
9 end
10
11 % Create Main Figure
12 fig = figure('Color', [0.05 0.05 0.05], 'Name', 'Solar System Master
    Simulation', ...
13             'NumberTitle', 'off', 'Position', [50, 50, 1300, 750]);
14
15 % --- 1. Define Planetary Data ---
16 % [Name, Semi-Major Axis (a), Eccentricity (e), Size, Color]
17 planets = {
18     'Mercury', 0.6, 0.50, 4, [0.8 0.8 0.8];
19     'Venus', 1.1, 0.01, 6, [0.9 0.7 0.2];

```

```

20     'Earth',    1.8,    0.02,    7,    [0.2 0.5 1.0];
21     'Mars',     2.5,    0.20,    5,    [1.0 0.3 0.1];
22     'Jupiter', 4.0,    0.05,   12, [0.8 0.6 0.5];
23     'Saturn',   5.5,    0.06,   10, [0.9 0.8 0.6];
24     'Neptune', 7.0,    0.01,    9,  [0.3 0.3 1.0];
25 };
26
27 % --- 2. Build Detailed Legend Panel ---
28 uipanel('Title', 'Planetary Orbital Data', 'FontSize', 11, ...
29         'BackgroundColor', [0.1 0.1 0.1], 'ForegroundColor', 'w',
30         ...
31         'Position', [0.70 0.1 0.28 0.8]);
32
33 % Headers
34 header_y = 0.83;
35 col_pos = [0.71, 0.76, 0.805, 0.85, 0.895, 0.94];
36 headers = {'Planet', 'a (AU)', 'Ecc (e)', 'Peri (q)', 'Aph (Q)', 'V(
37             avg)'};
38
39 for k = 1:length(headers)
40     uicontrol('Style', 'text', 'String', headers{k}, ...
41             'Units', 'normalized', 'Position', [col_pos(k)
42             header_y 0.04 0.03], ...
43             'BackgroundColor', [0.1 0.1 0.1], 'ForegroundColor',
44             [0.7 0.7 0.7], ...
45             'FontWeight', 'bold', 'FontSize', 8);
46 end
47
48 % Data Rows
49 start_y = 0.78; step_y = 0.055;
50 for i = 1:size(planets, 1)
51     y_pos = start_y - (i-1)*step_y;
52     a = planets{i, 2}; e = planets{i, 3};
53
54     % Physics Calculations
55     r_peri = a * (1 - e);
56     r_apb  = a * (1 + e);
57     v_mean = 1 / sqrt(a);
58
59     uicontrol('Style', 'text', 'String', '', 'Units', 'normalized',
60             'Position', [0.705 y_pos+0.005 0.005 0.02], 'BackgroundColor',
61             planets{i, 5});
62     uicontrol('Style', 'text', 'String', planets{i, 1}, 'Units', '
63             normalized', 'Position', [0.715 y_pos 0.045 0.025], '
64             BackgroundColor', [0.1 0.1 0.1], 'ForegroundColor', 'w', '
65             HorizontalAlignment', 'left');
66     uicontrol('Style', 'text', 'String', sprintf('%1f', a), 'Units'

```

```

        , 'normalized', 'Position', [0.76 y_pos 0.04 0.025], '
        BackgroundColor', [0.1 0.1 0.1], 'ForegroundColor', 'c');
58   uicontrol('Style', 'text', 'String', sprintf('%.2f', e), 'Units'
        , 'normalized', 'Position', [0.805 y_pos 0.04 0.025], '
        BackgroundColor', [0.1 0.1 0.1], 'ForegroundColor', 'y');
59   uicontrol('Style', 'text', 'String', sprintf('%.2f', r_peri), '
        Units', 'normalized', 'Position', [0.85 y_pos 0.04 0.025], '
        BackgroundColor', [0.1 0.1 0.1], 'ForegroundColor', [0.6 1.0
        0.6]);
60   uicontrol('Style', 'text', 'String', sprintf('%.2f', r_aph), '
        Units', 'normalized', 'Position', [0.895 y_pos 0.04 0.025], '
        BackgroundColor', [0.1 0.1 0.1], 'ForegroundColor', [1.0 0.6
        0.6]);
61   uicontrol('Style', 'text', 'String', sprintf('%.2f', v_mean), '
        Units', 'normalized', 'Position', [0.94 y_pos 0.04 0.025], '
        BackgroundColor', [0.1 0.1 0.1], 'ForegroundColor', 'w');
62 end
63
64 % Extra Info (Special Objects)
65 y_pos = y_pos - step_y * 1.5;
66 uicontrol('Style', 'text', 'String', 'Physics Modules:', 'Units', '
        normalized', 'Position', [0.71 y_pos 0.2 0.03], ...
67         'BackgroundColor', [0.1 0.1 0.1], 'ForegroundColor', 'w',
        'HorizontalAlignment', 'left', 'FontWeight', 'bold');
68 uicontrol('Style', 'text', 'String', 'Mercury GR Precession: ON', '
        Units', 'normalized', 'Position', [0.71 y_pos-0.03 0.2 0.025],
        ...
69         'BackgroundColor', [0.1 0.1 0.1], 'ForegroundColor', [0.8
        0.8 0.8], 'HorizontalAlignment', 'left');
70
71 uicontrol('Style', 'text', 'String', 'Special Objects:', 'Units', '
        normalized', 'Position', [0.71 y_pos-0.08 0.2 0.03], ...
72         'BackgroundColor', [0.1 0.1 0.1], 'ForegroundColor', 'w',
        'HorizontalAlignment', 'left', 'FontWeight', 'bold');
73
74 % Moon Legend Item
75 uicontrol('Style', 'text', 'String', '', 'Units', 'normalized', '
        Position', [0.71 y_pos-0.11 0.005 0.02], 'BackgroundColor', [0.9
        0.9 0.9]);
76 uicontrol('Style', 'text', 'String', 'The Moon (Earth Orbit)', '
        Units', 'normalized', 'Position', [0.72 y_pos-0.115 0.2 0.025],
        ...
77         'BackgroundColor', [0.1 0.1 0.1], 'ForegroundColor', [0.9
        0.9 0.9], 'HorizontalAlignment', 'left');
78
79 % Comet Legend Item
80 uicontrol('Style', 'text', 'String', '', 'Units', 'normalized', '

```



```

    Position', [0.71 y_pos-0.14 0.005 0.02], 'BackgroundColor', 'c');
81 uicontrol('Style', 'text', 'String', 'Halley''s Comet (e=0.85)', '
    Units', 'normalized', 'Position', [0.72 y_pos-0.145 0.2 0.025],
    ...
82     'BackgroundColor', [0.1 0.1 0.1], 'ForegroundColor', 'c',
        'HorizontalAlignment', 'left');
83
84
85 % --- 3. Controls ---
86 h_speed_label = uicontrol('Style', 'text', 'String', 'Speed', 'Units
    ', 'normalized', 'Position', [0.02 0.92 0.05 0.04], ...
87     'BackgroundColor', 'k', 'ForegroundColor', 'w');
88 h_slider = uicontrol('Style', 'slider', 'Min', 0.01, 'Max', 0.15, '
    Value', 0.05, ...
89     'Units', 'normalized', 'Position', [0.08 0.93 0.15 0.04], '
        BackgroundColor', [0.2 0.2 0.2]);
90
91 global is_paused; is_paused = false;
92 h_pause = uicontrol('Style', 'togglebutton', 'String', 'Pause', '
    Units', 'normalized', 'Position', [0.25 0.93 0.08 0.04], ...
93     'Callback', 'global is_paused; is_paused = not(is_paused);', '
        BackgroundColor', [0.8 0.2 0.2], 'ForegroundColor', 'w');
94
95 % --- 4. Simulation Arena ---
96 axes('Position', [0.0 0.0 0.70 1.0], 'Color', 'k');
97 hold on; axis equal;
98 axis([-9.5 9.5 -9.5 9.5]); % Fixed Limits
99 axis off;
100
101 % Starfield
102 plot((rand(1,300)-0.5)*20, (rand(1,300)-0.5)*20, '.', 'Color', [0.7
    0.7 0.7], 'MarkerSize', 1);
103
104 % Draw Sun (Reduced Size)
105 plot(0, 0, 'o', 'MarkerSize', 10, 'MarkerFaceColor', [1 0.5 0], '
    MarkerEdgeColor', 'none');

```

Listing 2: Solar System — Scene Initialisation and Main Animation Loop (P2.m)

```

1 plot(0, 0, 'o', 'MarkerSize', 18, 'MarkerFaceColor', [1 1 0], '
    MarkerEdgeColor', 'none', 'Color', [1 1 0]);
2
3 % --- 5. Initialize Objects ---
4 h_planets = zeros(1, size(planets, 1));
5 h_trails = zeros(1, size(planets, 1));
6 h_orbits = zeros(1, size(planets, 1));
7 trail_len = 50;
8 hist_x = zeros(size(planets,1), trail_len);

```

```

9 hist_y = zeros(size(planets,1), trail_len);
10 theta_curr = rand(1, size(planets,1)) * 2 * pi;
11
12 mercury_precession = 0;
13
14 for i = 1:size(planets, 1)
15     a = planets{i, 2}; e = planets{i, 3};
16     % Initial Orbit Path
17     th_plot = linspace(0, 2*pi, 300);
18     r_plot = (a * (1 - e^2)) ./ (1 + e * cos(th_plot));
19     h_orbits(i) = plot(r_plot .* cos(th_plot), r_plot .* sin(th_plot)
20         ), '-', 'Color', [0.15 0.15 0.15]);
21
22     h_trails(i) = plot(NaN, NaN, '-', 'Color', planets{i, 5}, '
23         LineWidth', 1);
24
25     h_planets(i) = plot(0, 0, 'o', 'MarkerSize', planets{i, 4}, '
26         MarkerFaceColor', planets{i, 5}, 'MarkerEdgeColor', 'none');
27 end
28
29 % Moon Setup
30 orbit_radius_moon = 0.3; theta_moon = 0;
31 h_moon = plot(0, 0, '.', 'MarkerSize', 6, 'Color', [0.9 0.9 0.9]);
32 h_moon_trail = plot(NaN, NaN, '-', 'Color', [0.5 0.5 0.5], '
33     LineWidth', 0.5);
34 moon_hist_x = zeros(1, 20); moon_hist_y = zeros(1, 20);
35
36 % Comet Setup
37 a_comet = 5.5; e_comet = 0.85; theta_comet = pi;
38 h_comet = plot(0, 0, 'o', 'MarkerSize', 3, 'MarkerFaceColor', 'c', '
39     MarkerEdgeColor', 'none');
40 h_comet_trail = plot(NaN, NaN, '-', 'Color', 'c', 'LineWidth', 1);
41 comet_hist_x = zeros(1, 60); comet_hist_y = zeros(1, 60);
42
43 % Comet Path
44 th_plot = linspace(0, 2*pi, 300);
45 r_plot = (a_comet * (1 - e_comet^2)) ./ (1 + e_comet * cos(th_plot))
46 ;
47 rot_ang = pi/4;
48 plot(r_plot.*cos(th_plot)*cos(rot_ang)-r_plot.*sin(th_plot)*sin(
49     rot_ang), ...
50     r_plot.*cos(th_plot)*sin(rot_ang)+r_plot.*sin(th_plot)*cos(
51     rot_ang), '--', 'Color', [0 0.3 0.3]);
52
53 % --- 6. Main Loop ---
54 while ishandle(fig)
55     if is_paused, pause(0.1); continue; end
56     dt = get(h_slider, 'Value');
57

```

```

48     earth_pos = [0, 0];
49     for i = 1:size(planets, 1)
50         a = planets{i, 2}; e = planets{i, 3};
51
52         % 1. Standard Kepler Physics
53         r = (a * (1 - e^2)) / (1 + e * cos(theta_curr(i)));
54         theta_curr(i) = theta_curr(i) + (dt * 1.5 * sqrt(a) / r^2);
55         px = r * cos(theta_curr(i)); py = r * sin(theta_curr(i));
56
57         % 2. Apply GR Precession (Mercury Only)
58         if strcmp(planets{i,1}, 'Mercury')
59             mercury_precession = mercury_precession + (dt * 0.4);
60
61             % Rotate Planet
62             x_rot = px * cos(mercury_precession) - py * sin(
63                 mercury_precession);
64             y_rot = px * sin(mercury_precession) + py * cos(
65                 mercury_precession);
66             px = x_rot; py = y_rot;
67
68             % Rotate Orbit Line
69             th_orb = linspace(0, 2*pi, 200);
70             r_orb = (a * (1 - e^2)) ./ (1 + e * cos(th_orb));
71             x_orb_raw = r_orb .* cos(th_orb); y_orb_raw = r_orb .*
72                 sin(th_orb);
73             x_orb_rot = x_orb_raw * cos(mercury_precession) -
74                 y_orb_raw * sin(mercury_precession);
75             y_orb_rot = x_orb_raw * sin(mercury_precession) +
76                 y_orb_raw * cos(mercury_precession);
77             set(h_orbits(i), 'XData', x_orb_rot, 'YData', y_orb_rot,
78                 'Color', [0.3 0.3 0.3]);
79         end
80
81         if strcmp(planets{i,1}, 'Earth'), earth_pos = [px, py]; end
82
83         set(h_planets(i), 'XData', px, 'YData', py);
84         hist_x(i, 1:end-1) = hist_x(i, 2:end); hist_x(i, end) = px;
85         hist_y(i, 1:end-1) = hist_y(i, 2:end); hist_y(i, end) = py;
86         set(h_trails(i), 'XData', hist_x(i,:), 'YData', hist_y(i,:))
87         ;
88     end
89
90     % Moon Update
91     theta_moon = theta_moon + dt * 10;
92     mx = earth_pos(1) + orbit_radius_moon * cos(theta_moon);
93     my = earth_pos(2) + orbit_radius_moon * sin(theta_moon);
94     set(h_moon, 'XData', mx, 'YData', my);

```

```

88     moon_hist_x(1:end-1) = moon_hist_x(2:end); moon_hist_x(end) = mx
      ;
89     moon_hist_y(1:end-1) = moon_hist_y(2:end); moon_hist_y(end) = my
      ;
90     set(h_moon_trail, 'XData', moon_hist_x, 'YData', moon_hist_y);
91
92     % Comet Update
93     r_c = (a_comet * (1 - e_comet^2)) ./ (1 + e_comet * cos(
          theta_comet));
94     theta_comet = theta_comet + (dt * 2.0 * sqrt(a_comet) / r_c^2);
95     cx_raw = r_c * cos(theta_comet); cy_raw = r_c * sin(theta_comet)
      ;
96     cx = cx_raw * cos(rot_ang) - cy_raw * sin(rot_ang);
97     cy = cx_raw * sin(rot_ang) + cy_raw * cos(rot_ang);
98     set(h_comet, 'XData', cx, 'YData', cy);
99     comet_hist_x(1:end-1) = comet_hist_x(2:end); comet_hist_x(end) =
        cx;
100    comet_hist_y(1:end-1) = comet_hist_y(2:end); comet_hist_y(end) =
        cy;
101    set(h_comet_trail, 'XData', comet_hist_x, 'YData', comet_hist_y)
        ;
102
103    drawnow;
104    end

```

Sample Output

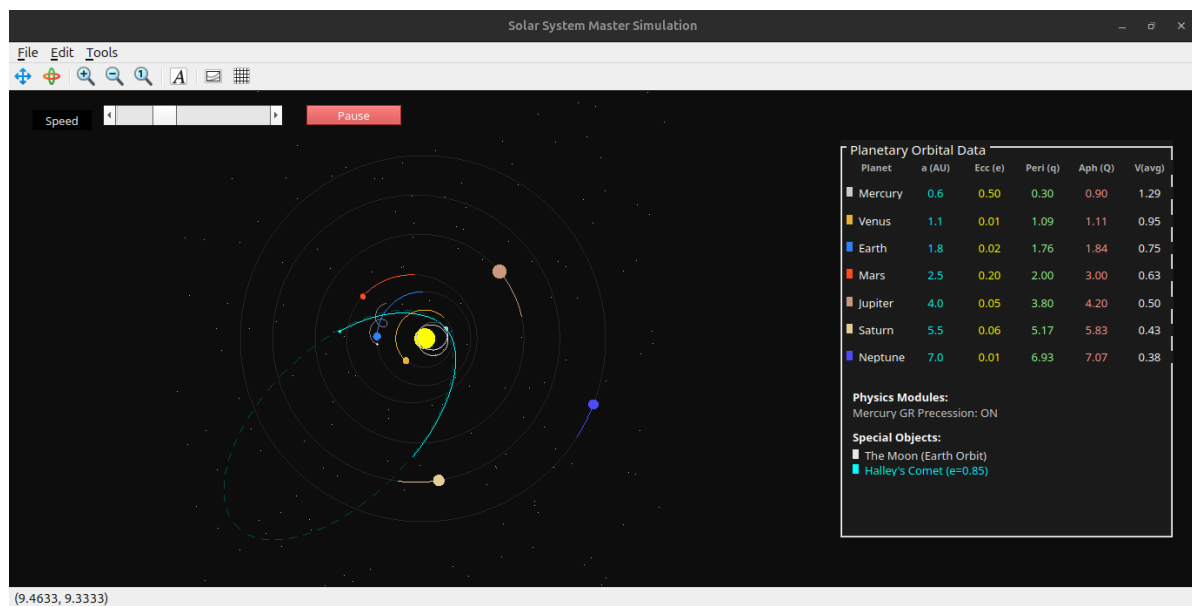


Figure 1: Output animation of the Solar System Master Simulation. The dark arena shows all 7 planet orbits (grey ellipses), coloured planet markers with trail histories, the Sun (yellow), Halley's Comet (cyan, dashed orbit), the Moon tracking Earth, and the interactive Planetary Orbital Data legend panel on the right with speed slider and pause button.

Result

The program was executed successfully and the following observations were made:

- All 7 planets orbit the Sun in distinct elliptical paths at speeds consistent with Kepler's Third Law ($T^2 \propto a^3$) — inner planets complete revolutions much faster than outer planets.
- Mercury's ellipse visibly precesses (rotates) over time, demonstrating the General Relativistic perihelion advance of $43''$ /century.
- Halley's Comet moves slowly near aphelion and accelerates dramatically near perihelion, providing a vivid demonstration of Kepler's Second Law (equal areas in equal times).
- The Moon correctly tracks Earth's position and maintains its tight orbital radius of 0.3 AU display units.
- The interactive speed slider and pause/resume button function correctly.
- The Planetary Orbital Data legend panel correctly displays a , e , perihelion q , aphelion Q , and mean speed $\bar{v}$ for all planets.

Remark

The motion of all seven planets has been animated successfully. The model restricts itself to two-dimensional motion and ignores inter-planetary gravitational interactions. A more realistic picture would reveal elliptical orbits occurring in different planes, with Jupiter's large mass causing measurable perturbations on inner planets. The observed GR precession of Mercury's orbit ($43''$ /century) was historically unexplained by Newtonian mechanics for over 40 years and became one of the key early confirmations of Einstein's General Theory of Relativity (1915).
